QNH MANUAL FOR OPERATING AI DEBUGGER

INTRODUCTION

Thank you all for cloning this GitHub repo, this manual helps you to understand and operate all the functionalities needed to run this AI

ENV FILE

- The env file or more famously known as the environment file was locked in ".gitignore", and hence was not uploaded to git repo
- You need to create a google gemini API key for this to function. Go to link : https://aistudio.google.com/welcome?gad_campaignid=23417416052 or search for "google gemini key" and follow the website
- Paste the api key in your .env file, and the file itself should be present inside your working folder

Model Sim

- Model sim is an Intel FPGA software, and can be found on intel's website. Go to link : https://www.intel.com/content/www/us/en/software-kit/868561/intel-quartus-prime-lite-edition-design-software-version-25-1-for-windows.html or follow YouTube tutorial for installation, YouTube link -> https://youtu.be/yRipAryv0A8

Python

- Python is required. Current uses python version 3.14.2
- If you are from future, its advised to create a python venv, and downgrade it to the abovementioned python version

PIP REQUIREMENTS

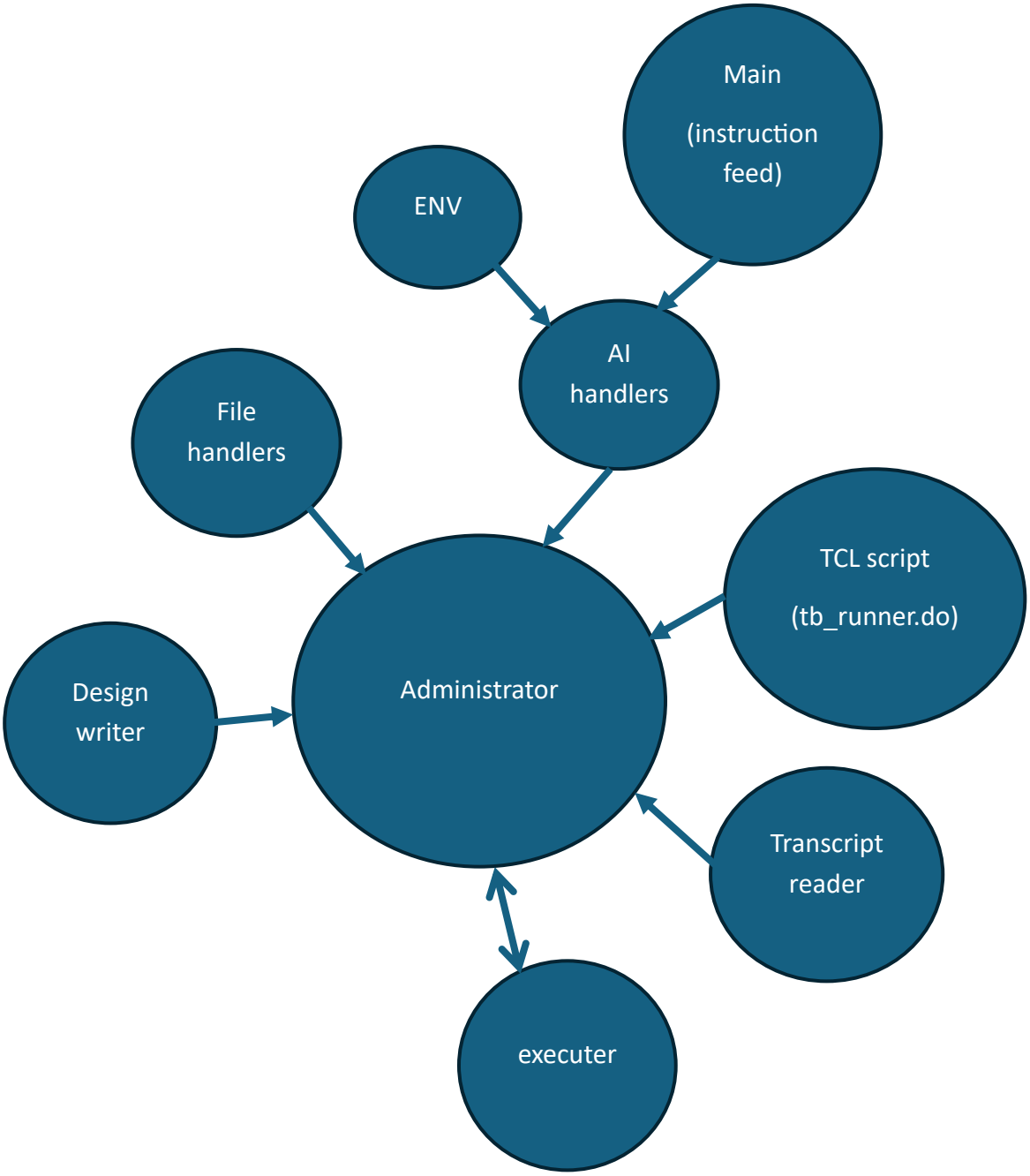
- The pip required are mentioned in the file "install pips". Its strongly advised to use the same the library version, if python version > 3.14

SYSTEM VERILOG KNOWLEDGE

- The AI is made to generate the testbench for RTL design, and not to generate the design itself. Hence with all my heart, I request you to know the knowledge of systemverilog before starting this, and also tcl script

WORKFLOW

The graph below shows the workflow of python, and can be useful for any debugging, push requests to improve the software. This gentlemen is the V0 of the AI debugger


Main
(instruction
feed)
ENV
AI
handlers
File
handlers
TCL script
(tb_runner.do)
Design
writer
Administrator
Transcript
reader
executer

File handler

The file handler has 3 functions :
tb_path, design_path, and test_plan_path

- tb_path : This function must have the relative path to your testbench
- design_path : This function must have the relative path to your design, although this is optional
- test_plan_path : This function is reserved and is in development. This function connects to nowhere

Design Writer

The file has 1 function :

- design_writer : this function is a buffer to the gemini api write, and is an atomic write. I recommend not to modify this function. If you are a developer, who knows what he is doing, you can improve the file, and submit a push request
- The file also has protection against illegal writes from gemini

Transcript Reader

The file has 1 function :

- get_final_error_count : This function uses the python's native library "re", to count the "error" marked by systemverilog automated tasks.

TCL script

The file is not a python function, and hence is called using OS

==========WARNING TO MAC AND LINUX USERS==========

This function might not work in mac and linux due to OS compatibility, and hence, follow alternative methods

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For windows users:

There is no need to edit any files, other than the required parts

For mac and linux :

You are instructed to edit administrator.py and main.py, as both of them use os.

However, in the age of AI, this shouldn't be an issue

The TCL script invokes the modelSim software to run. Knowledge on modelSim and tcl is encouraged

ENV

This is a very important file, and should start by GOOGLE_API_KEY = XXXXXXXXXXXX

Where "XXXXXXXXXXXX", is your generated API key

main

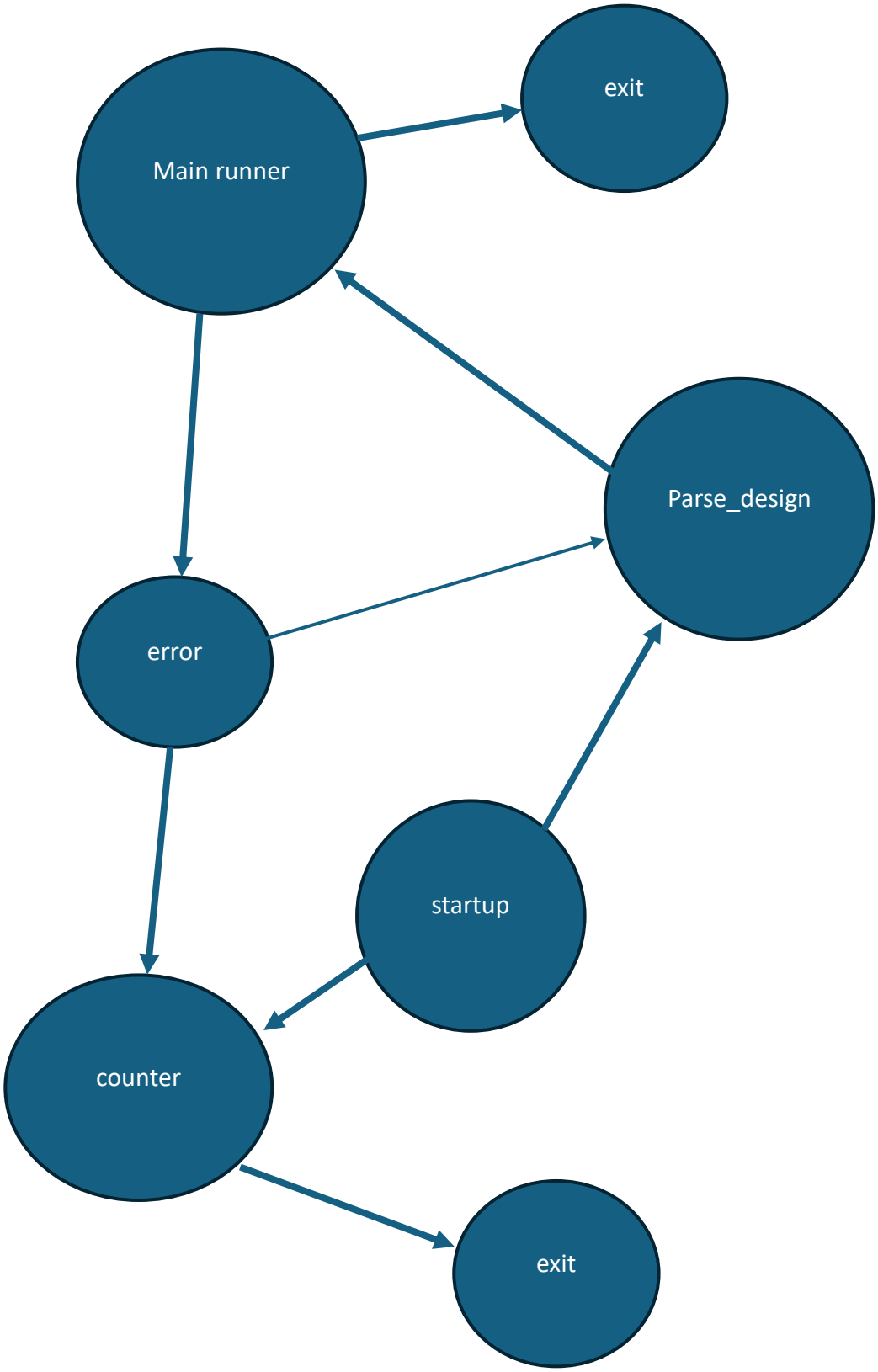
This is not the main software, be warned, but it's the first milestone. This file has 5 functions. Mac and linux users, be warned that this files uses os functionality

- main : This is the pdf for the test plan, and is reserved, you are not supposed to invoke this function. You are warned
- dump : This function instruction gemini to provide a vcd dump file feature to your testbench
- end_statement : You are instructed not to modify the function unless needed. This function strictly instructs gemini to not to use any sort of em-dashes, or any other symbols, which are out of syntax
- plans : You must edit this function according to your design, and you need to present the plan for your design such as the pin-out the bits any dependencies such as pkg files, .vh files used
- combiner : By default, the combiner does not invoke the vcd dump files

AI handler

AI handler has 2 functions :

- AI_handler : this function takes an input of text, and needs to be provided. You are advised to run this function once independently, after getting the API key. This function is used only to test weather API key is working
- AI_handler_with_input : this function has a default input. You are advised not to change the default

ADMINSTRATOR

This is the workhorse of the entire program and runs in a FSM fashion
You are in final control in running this, and need to input “F” in the commandline when prompted